

One-step lookahead in safety-gated dual-agent dose-finding: an operating-characteristic study and a decision rule

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MAIN RESULT: one-step lookahead buys recommended-dose efficacy - not a universal improvement in localization or safety.

1. Clinical question / why pharma should care

Early-phase combination trials must identify an optimal biological dose (OBD): the most efficacious dose pair whose toxicity remains acceptable.

- Final recommendation efficacy, OBD localization, recommendation safety, and in-trial exposure are different operating characteristics.
- We hold the surrogate, gate, grid, budget, and recommendation rule fixed; only the next-cohort acquisition changes.
- Project Optimus context: optimize dosage, not simply the maximum tolerated dose.

2. Safety-gated dose-finding setup

$$d^*(z) = \arg \max_{d \in [0,1]^2} e(d, z) \quad \text{s.t.} \quad g(d, z) \leq g^\dagger(z)$$

$$\text{PF}(d) = \Phi\left(\frac{g^\dagger - \mu_g(d)}{\sigma_g(d)}\right), \quad A_\tau = \{d : \text{PF}(d) > \tau\}$$

$$d_{\text{next}} = \arg \max_{d \in A_\tau \cap G} \alpha(d), \quad \hat{d} = \arg \max_{d \in A_\tau} \mu_e(d)$$

Benchmark design

2 agents; 2 strata 5 x 5 grid N = 40 / stratum cohort = 2 4 warm-up

Gate sweep: tau = 0.5, 0.7, 0.9 | No early stopping | OSA: n = 200/cell | Logistic and MARIPOSA: n = 100

Compared policies (everything else fixed)

cEI	Myopic expected improvement x posterior feasibility.
cKG	One-step expected gain in the gated best efficacy.
cEI-tMSE	Alternate cEI with targeted boundary learning.

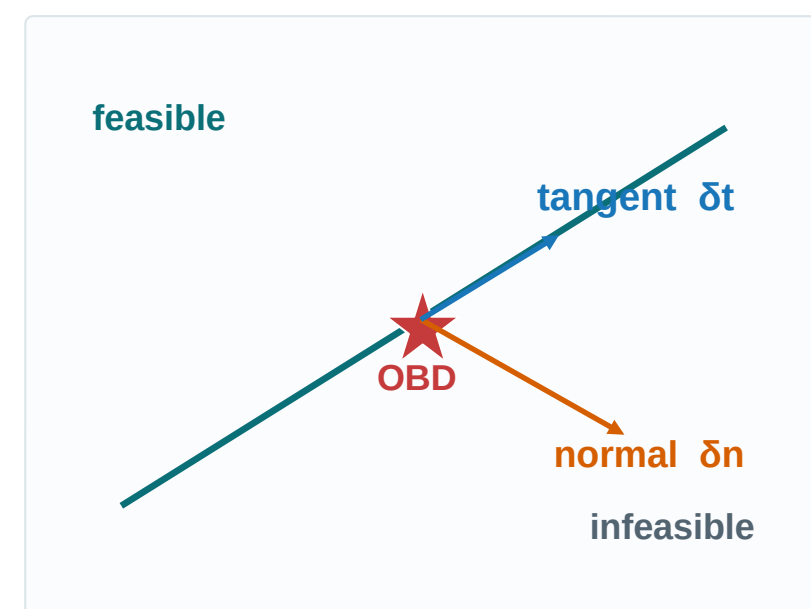
3. Why lookahead concentrates at a binding boundary

At a boundary dose, myopic cEI is blind to toxicity-posterior uncertainty; a strict-gate cKG is not.

$$\frac{\partial \alpha_{\text{cEI}}}{\partial \sigma_g} = 0 \quad (\mu_g = g^\dagger) \quad \text{At the boundary, the cEI feasibility factor is always 1/2.}$$

$$\frac{\alpha_{\text{cKG}}(d)}{h-b} = \Phi\left[-\Phi^{-1}(\tau) \frac{\sigma_{y_g}}{\sigma_g}\right], \quad \tau > \frac{1}{2} \quad \text{Strict-gate value of information increases with posterior toxicity SD.}$$

Constraint-binding geometry at the OBD



$$V(d^* + \delta_n) - V(d^*) = -\lambda \|\nabla g\| |\delta_n| + O(\delta_n^2)$$

Across the boundary: first-order value loss

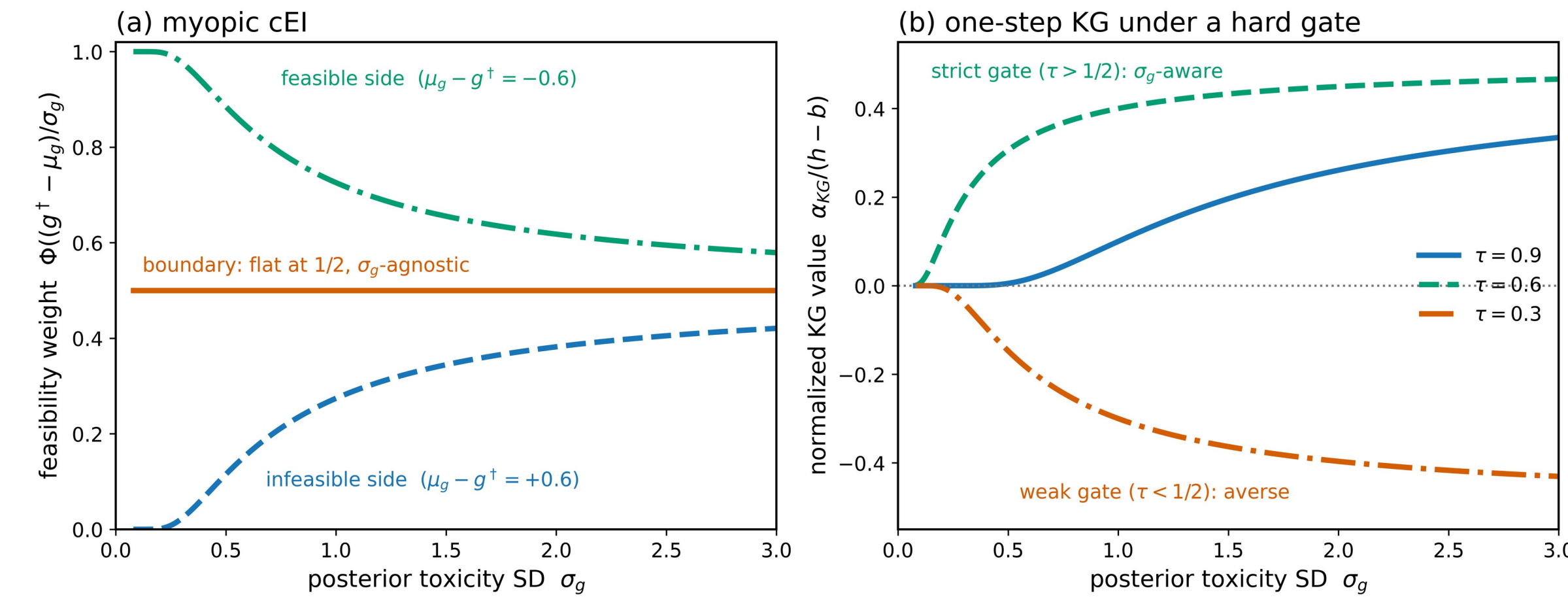
$$V(d^* + \delta_t) - V(d^*) = -\frac{1}{2} H_{\tan} \delta_t^2 + o(\delta_t^2)$$

Along the boundary: second-order value loss

A value-greedy lookahead rule is pulled toward across-boundary uncertainty. It can improve recommendation value without minimizing dose distance or recommendation risk.

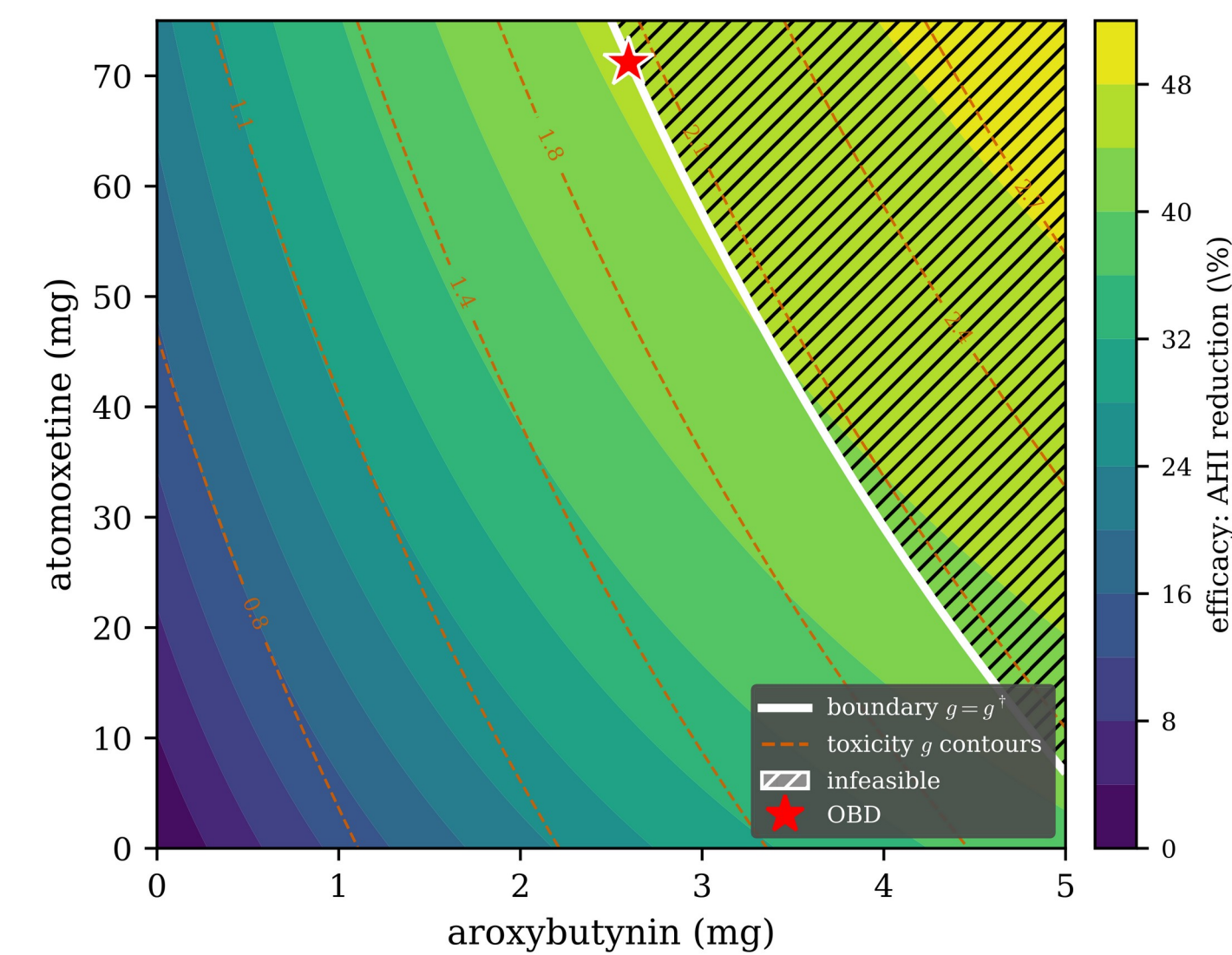
Scope: a directional mechanism at a regular binding optimum - not a claim of universal finite-sample dominance.

4. Boundary mechanism (adapted from paper Fig. 1)



cEI has no first-order incentive to sharpen boundary uncertainty; cKG values an observation that may admit a high-efficacy boundary dose.

5. Clinically anchored surface: MARIPOSA-informed calibration



Original paper Fig. 2

Clinical anchor

- Efficacy: percent AHI reduction.
- Toxicity: antimuscarinic adverse-event burden increasing with aroxybutynin.
- Two-drug optimum lies on the tolerability boundary.

Calibrated OBD

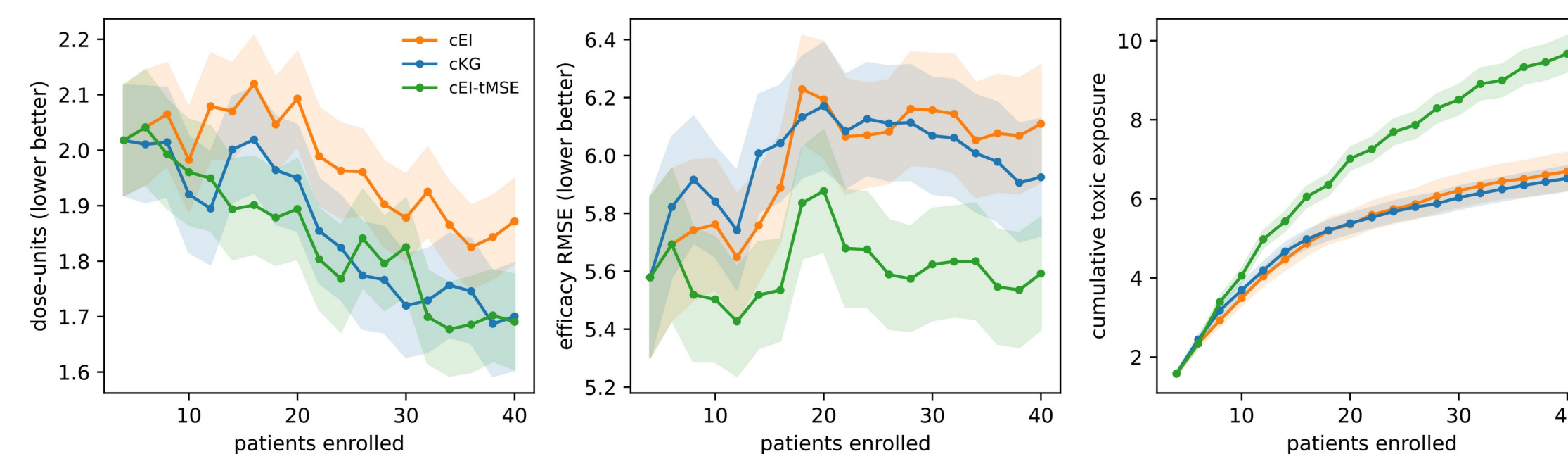
2.6 mg aroxybutynin
71 mg atomoxetine
45% AHI reduction

Matches by construction the 2.5/75 mg dose selected in phase 2b MARIPOSA.

Interpretation boundary

Aggregate calibration only. No patient-level data. This is a clinically anchored consistency check - not independent clinical validation.

6. What happens during enrollment? OSA, tau = 0.7



Original paper Fig. 3: 80 paired seed labels pooled over two strata; bands are +/-1 seed-clustered Monte Carlo SE.

At N = 40: dose-units = 1.87 +/- 0.08 (cEI), 1.70 +/- 0.10 (cKG), 1.69 +/- 0.09 (cEI-tMSE). Cumulative truly-toxic assignments = 6.69 +/- 0.48, 6.52 +/- 0.31, 9.67 +/- 0.45.

7. Main operating characteristics at tau = 0.7

Means shown; lower is better except gate precision and efficacy gain.

Metric / surface	cEI	cKG	cEI-tMSE
OSA unsafe recommendation % (z=0 / z=1)	13.5 / 6.0	23.0 / 21.0	7.0 / 6.5
OSA gate precision % (z=0 / z=1)	76.2 / 82.1	87.8 / 90.8	96.2 / 95.2
Logistic dose-units / unsafe %	1.18 / 10.5	1.32 / 22.5	1.15 / 10.5
MARIPOSA dose-units / unsafe %	1.90 / 0.0	1.54 / 21.0	1.63 / 10.0

Multiplicity-aware location result

- cKG beat cEI in 3 of 6 OSA cells after Holm correction under the continuous binding target - but only 1 of 6 after re-scoring to the source-panel OBD.
- Against cEI-tMSE, cKG has no general localization advantage; the logistic surface favors lookahead-free rules.

Recommended-dose efficacy gain: cKG minus cEI

OSA +0.872 +/- 0.223 surface efficacy units	Logistic +0.029 +/- 0.006 surface efficacy units	MARIPOSA +4.349 +/- 0.716 percentage-point scale
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8. Objective-specific decision guide for pharma

Highest final efficacy	cKG	Highest mean true efficacy at tau = 0.7 on all four tested surfaces.
Recommendation safety	cEI or cEI-tMSE	Avoid cKG boundary aggression; ordering of the lookahead-free rules is surface-dependent.
OBD localization	Simulate cEI and cEI-tMSE	No general cKG advantage over cEI-tMSE; logistic results favor lookahead-free rules.
In-trial exposure	Evaluate directly	Exposure ordering differs from final recommendation safety.
Stricter gate	Any candidate policy	Reduces cKG unsafe recommendations, but does not guarantee best localization or lowest exposure.

9. Clinical translation: what is still missing

Current evidence	Before deployment
- Acquisition-policy operating characteristics - not a whole trial design. - Continuous latent efficacy/toxicity; fixed N; posterior chance gate; no hard elimination/stopping. - Aggregate clinical anchor; no patient-level validation.	- Overdose-probability cap, dose elimination/stopping, and a no-dose option. - Binary/delayed outcomes and late-onset toxicity handling. - Calibrated GP uncertainty, prior sensitivity/data-conflict safeguards. - Clinical utility elicitation, regulatory error metrics, validated software.

10. Take-home message

Choose and validate the acquisition for the prioritized clinical objective. Report final efficacy, localization, recommendation safety, and in-trial exposure separately.